

CIRIK-RABAT, Kazakhstan

WMO: 380490

Lat: 44.067N

Lon: 62.900E

Elev: 289

StdP: 14.54

Time Zone: 5.00 (E05)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.5	-1.8	-14.5	2.5	-4.8	-9.2	3.4	-1.3	27.3	19.0	23.1	13.3	2.6	20	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	103.4	72.7	100.3	70.9	97.4	69.9	78.9	96.1	75.7	93.3	73.1	91.6	5.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	125.5	93.0	69.7	110.3	88.5	66.3	98.0	86.5	42.7	95.9	39.4	93.0	36.9	91.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	17.8	15.5	DB	-10.0	109.6	8.7	3.8	-16.2	112.3	-21.3	114.5	-26.2	116.6	-32.6	119.4	(4)
(5)			WB	-10.4	79.3	8.6	6.0	-16.6	83.6	-21.7	87.1	-26.5	90.5	-32.8	94.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	22.0	21.7	35.8	55.8	68.0	78.9	84.0	78.4	65.6	50.2	34.8	25.2	(6)	
	DBStd	24.13	10.90	11.91	10.21	9.20	7.88	6.87	5.58	6.10	7.29	9.16	10.69	11.09	(7)	
	HDD50	3508	868	794	453	47	3	0	0	0	2	107	464	770	(8)	
	HDD65	6500	1333	1214	906	299	60	3	0	1	78	465	906	1235	(9)	
	CDD50	4186	0	0	12	220	561	868	1053	880	472	113	7	0	(10)	
	CDD65	1703	0	0	0	21	153	421	588	416	98	6	0	0	(11)	
	CDH74	24061	0	0	1	295	1967	5996	8867	5524	1297	114	0	0	(12)	
	CDH80	13266	0	0	0	78	800	3395	5462	3010	496	25	0	0	(13)	
	Wind	WSAvg	4.8	4.0	5.3	5.8	6.0	5.0	5.3	5.2	4.7	4.3	3.7	4.2	3.8	(14)
	Precipitation	PrecAvg	7.0	1.0	1.0	1.1	0.7	0.6	0.3	0.1	0.1	0.0	0.3	0.8	0.9	(15)
(16) Precipitation	PrecMax	11.6	2.6	2.4	2.4	2.2	2.3	1.4	0.7	0.8	0.3	1.0	2.3	2.6	(16)	
	PrecMin	3.9	0.2	0.2	0.3	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)	
	PrecStd	2.0	0.5	0.6	0.5	0.5	0.6	0.4	0.2	0.2	0.1	0.2	0.6	0.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.9	54.3	68.7	87.7	96.0	105.7	108.3	103.8	95.5	83.3	67.0	51.4	(19)	
		MCWB	42.3	45.3	53.0	65.4	67.3	75.3	75.5	74.4	67.3	61.7	52.7	43.1	(20)	
	2%	DB	42.1	44.6	61.9	81.4	91.4	101.6	105.0	100.1	89.7	76.3	58.8	44.8	(21)	
		MCWB	37.6	40.6	49.5	61.7	65.7	70.9	72.7	70.9	64.6	57.6	47.9	39.1	(22)	
	5%	DB	37.1	39.3	56.3	77.1	87.8	98.0	101.8	96.8	85.9	71.5	53.8	40.5	(23)	
		MCWB	33.8	35.3	47.0	58.9	63.7	70.0	71.4	69.4	62.9	55.1	45.3	36.5	(24)	
	10%	DB	34.5	35.9	50.6	72.6	84.3	94.7	98.8	93.5	81.7	66.3	49.0	37.8	(25)	
		MCWB	31.8	32.5	43.5	56.6	62.9	69.0	70.9	67.7	60.8	51.7	42.9	34.6	(26)	
	0.4%	WB	42.5	47.4	53.8	67.7	73.0	80.8	85.2	81.9	73.3	63.3	54.0	43.4	(27)	
		MCDB	48.4	53.1	67.7	84.6	87.3	99.9	103.8	98.1	89.5	79.4	64.2	49.2	(28)	
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.7	40.8	50.9	63.4	69.4	76.2	80.1	76.7	68.8	59.4	49.7	39.8	(29)	
		MCDB	42.0	44.1	60.1	78.4	85.8	93.8	97.2	94.3	83.7	74.7	56.3	44.2	(30)	
	5%	WB	34.4	35.8	47.7	60.6	66.3	73.7	76.9	73.4	64.6	55.8	46.4	37.3	(31)	
		MCDB	36.7	39.0	56.1	74.9	83.9	92.6	94.2	91.0	81.7	68.9	52.8	40.2	(32)	
	10%	WB	32.1	32.7	43.9	57.5	63.9	71.2	73.9	70.2	61.6	52.5	42.9	34.9	(33)	
		MCDB	34.1	35.6	50.2	70.7	82.0	90.9	92.4	88.7	79.4	65.2	48.6	37.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	15.4	16.7	23.1	24.8	25.7	25.3	25.4	26.2	24.1	16.6	12.9	(35)	
		MCDBR	13.8	17.2	25.7	28.7	29.5	29.2	29.4	29.7	31.3	30.3	24.3	16.7	(36)	
		MCWBR	10.9	12.7	15.5	16.0	14.5	14.4	13.5	14.7	16.7	18.1	16.1	13.1	(37)	
	5% WB	MCDBR	12.7	17.2	23.9	27.0	27.7	26.6	26.7	27.6	28.7	29.4	22.1	15.2	(38)	
		MCWBR	10.4	13.1	15.0	16.4	16.7	16.2	16.4	17.7	17.8	18.3	15.2	12.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.362	0.391	0.449	0.493	0.495	0.477	0.456	0.440	0.409	0.405	0.385	0.365	(40)	
	taud		2.075	2.021	1.917	1.844	1.888	1.974	2.057	2.073	2.105	2.055	2.079	2.071	(41)	
	Ebn at Noon		226	243	244	245	249	254	258	258	255	234	215	209	(42)	
	Edh at Noon		33	42	53	62	61	56	51	49	43	40	33	30	(43)	
(44) All-Sky Solar Radiation	RadAvg		473	816	1249	1728	2181	2395	2334	2126	1681	1099	634	418	(44)	
	RadStd		71	125	112	131	120	104	85	70	59	62	70	57	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	-2.29	N/A	N/A	N/A	+142	

Nomenclature: See separate page